

AI BASED UNIFIED FRAMEWORK FOR SPEAKER IDENTIFICATION AND VERIFICATION IN ASSAMESE LANGUAGE

A Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of
MASTER OF COMPUTER APPLICATIONS (MCA)

TEZPUR UNIVERSITY

2026



Submitted by

KRISHNAMONI DAS (CSM24051)
PRIYANKU BORAH (CSM24058)

Guided by

DR. SANGHAMITRA NATH
ASSISTANT PROFESSOR
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Tezpur University
Tezpur – 784028, Assam



DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING
TEZPUR UNIVERSITY
TEZPUR – 784028, ASSAM, INDIA

CERTIFICATE

This dissertation entitled "**AI Based Unified Framework For Speaker Identification and Verification in Assamese Language**" submitted by **Mr. Krishnamoni Das** and **Mr. Priyanku Borah** in partial fulfilment of the requirements for the degree of **Master of Computer Application (MCA)** of Tezpur University has been examined.

Examiner

Examiner

Place:

Place:

Date:

Date:



TEZPUR UNIVERSITY

CERTIFICATE

This serves to confirm that the project report titled, "**AI Based Unified Framework For Speaker Identification and Verification in Assamese Language**" has been submitted by **Mr. Krishnamoni Das (CSM24051)** and **Mr. Priyanku Borah (CSM24058)**.

Both students have satisfactorily completed the project requirements, fulfilling the necessary conditions and adhering to the regulations for the conferral of the **Master of Computer Application** degree at **Tezpur University** during the academic session 2024–2026.

To the best of our understanding, the content presented has not been presented for the attainment of any Degree or Diploma at any other university or institute.

Place: Tezpur

Date: 02 June 2026

(Bhabesh Nath)

Professor & Head

Dept. of Computer Sc. & Engg.



TEZPUR UNIVERSITY

CERTIFICATE

This is to certify that the project report entitled, "**AI Based Unified Framework For Speaker Identification and Verification in Assamese Language**" submitted to the **Department of Computer Science and Engineering, Tezpur University**, in partial fulfilment for the award of the degree of **Master of Computer Application** in Computer Science and Engineering, is a record of bona fide work carried out by **Mr. Krishnamoni Das**, Roll No. **CSM24051** and **Mr. Priyanku Borah**, Roll No. **CSM24058**, under my supervision and guidance. All help received by them from various sources has been duly acknowledged. No part of this report has been submitted elsewhere for the award of any other degree.

Place: Tezpur

Date: 02 June 2026

(Dr.Sanghamitra Nath)

Assistant Professor

Supervisor

DECLARATION

We affirm that the project work entitled, "**AI Based Unified Framework For Speaker Identification and Verification in Assamese Language**" submitted to the Department of Computer Science and Engineering, Tezpur University, in partial fulfilment for the award of the degree of **Master of Computer Application** in Computer Science and Engineering, is a record of bona fide work carried out by **Mr. Krishnamoni Das** (Roll No. CSM24051) and **Mr. Priyanku Borah** (Roll No. CSM24058) under the supervision and guidance of **Dr. Sanghamitra Nath**.

We affirm that the project work entitled, "**AI Based Unified Framework For Speaker Identification and Verification in Assamese Language**" submitted to the **Department of Computer Science & Engineering at Tezpur University**, was authored solely by us and has not been presented to any other institution for the purpose of obtaining any other degree.

Place: Tezpur

Krishnamoni Das (CSM24051)

Date: 02 June 2026

Priyanku Borah (CSM24058)

ACKNOWLEDGEMENTS

We humbly express our deepest gratitude to **God Almighty** for His divine blessings, strength and guidance that enabled us to successfully accomplish this project.

We are profoundly indebted to our esteemed mentor, **Dr. Sanghamitra Nath**, Assistant Professor, Department of Computer Science and Engineering, Tezpur University, for her constant encouragement, insightful guidance and unwavering support throughout the course of this work.

We also extend our sincere thanks to Narottam Bhattacharjee, Research Scholar, Tezpur University, for his valuable suggestions, technical expertise and kind cooperation, which greatly contributed to the successful completion of this project.

We would like to acknowledge with gratitude all the individuals who contributed to the **dataset creation**. Their time, effort and willingness to participate played a vital role in making this work possible.

We are thankful to the faculty members of the Department of Computer Science and Engineering, Tezpur University, for providing the necessary academic environment and resources.

Finally, we express our heartfelt appreciation to our **parents, loved ones and friends** for their unwavering support, encouragement and belief in us, which served as a constant source of motivation throughout this journey.

Place: Tezpur

Krishnamoni Das (CSM24051)

Date: 02 June 2026

Priyanku Borah (CSM24058)

ABSTRACT

Speaker Identification and Speaker Verification are important voice-based biometric techniques used to recognize or authenticate a person from their speech. As voice-enabled systems and security applications are becoming more common, developing reliable speaker recognition systems has become an important area of research.

However, for the Assamese language, this field is still not well explored. Assamese is spoken by more than 15 million people in northeastern India, but it remains under-resourced in speech and speaker recognition research. There are very few publicly available Assamese speech datasets, and limited research work has been carried out in this area. This creates a strong need for developing speaker recognition systems specifically for the Assamese language.

This project presents the design and implementation of a **Speaker Identification and Verification system** for the **Assamese language** using both **Machine Learning** and **Deep Learning approaches**. Traditional machine learning classifiers - **Support Vector Machine (SVM)**, **Random Forest** and **Gradient Boosting** - are trained on **Mel-Frequency Cepstral Coefficients (MFCCs)** combined with spectral features. State of the art deep learning models - **X-Vector** and **ECAPA-TDNN** - extract robust speaker embeddings from log-Mel spectrograms.

A custom Assamese speech dataset was created from 20 speakers and preprocessed through silence removal, normalisation, Praat-based annotation, automated and manual trimming. The system supports closed-set speaker identification and binary verification using probability thresholds (ML) and cosine similarity on embeddings (DL). A Streamlit - based interactive web application allows users to select models and perform real-time speaker recognition.

For evaluation, identification performance is measured using accuracy, precision, recall, and F1-score. In addition, real-time identification and verification are analysed using confidence and similarity scores. Machine learning models provide probability-based confidence values, while deep learning models use cosine similarity between embeddings to determine speaker match.

An interactive web application built using Streamlit enables users to perform speaker identification and verification through audio upload or live recording. Experimental results show that deep learning models, particularly ECAPA-TDNN, achieve superior performance, while machine learning models provide strong baseline results. This work establishes an initial benchmark for Assamese speaker recognition and addresses the lack of existing datasets and evaluation frameworks for this language.

CONTENTS

ACKNOWLEDGEMENTS	VI
ABSTRACT	VII
LIST OF FIGURES	X
LIST OF TABLES	XI
LIST OF SYMBOLS AND ABBREVIATIONS	XII
INTRODUCTION	1
LITERATURE REVIEW	3
2.1 Classical and Statistical Approaches	3
2.2 Machine Learning Classifiers	4
2.3 Deep Learning and Speaker Embeddings	4
2.4 Speaker Recognition for Indian and Assamese Languages	5
2.5 Summary Table of Related Works	5
INITIAL SYSTEM STUDY	8
3.1 The Existing System	8
3.2 Drawbacks of the Existing System	8
3.3 Problem Definition	9
3.4 The Proposed System	10
3.5 Scope of the System	11
3.6 Scope of this Project	11
SOFTWARE AND HARDWARE REQUIREMENTS	12
4.1 Integrated Development Environment (IDE)	12
4.2 Programming Language	12
4.3 Libraries and Frameworks	12
4.4 Operating System	12
4.5 Hardware Requirements	13
SYSTEM DESIGN	14
5.1 Work Flow Diagram	14
IMPLEMENTATION	19
6.1 Speech Dataset Creation	19
6.2 Speech Annotation Using Praat	20
6.3 Automated Audio Trimming Using Python	23
6.4 Audio Preprocessing	24
6.5 Feature Extraction	25
6.6 Model Training	26
6.7 Model Evaluation and Performance Analysis	28
6.8 Models Used and Justification	30
6.9 Web-Based Application Development	31
6.10 System Integration and Testing	35
RESULTS	37
7.1 Identification Performance	37
7.2 Speaker 2×2 Confusion Matrix	38
7.3 Speaker-wise Confidence	41
7.4 Speaker-wise Precision / Recall / F1 Score Of Different Models	42

CONCLUSION AND FUTURE WORK	45
8.1 Conclusion	45
8.2 Future Work	45
References	47
Turnitin Similarity Report.....	52

LIST OF FIGURES

Figure 1: Work Flow Diagram.....	14
Figure 2: Praat Annotation - Read from File	21
Figure 3: Praat Annotation - TextGrid Dialog.....	22
Figure 4: Praat Annotation - Tier Structure	23
Figure 5: Accuracy Formula	28
Figure 6: Precision Formula.....	28
Figure 7: Recall Formula	29
Figure 8: F1-Score Formula.....	29
Figure 9: Support Formula.....	30
Figure 10: Streamlit Web Interface.....	32
Figure 11: Model Selection - Machine Learning.....	32
Figure 12: Model Selection - Deep Learning	33
Figure 13: Select Voice.....	34
Figure 14: Voice loaded.....	34
Figure 15: Predicted Result - Identification.....	34
Figure 16: Predicted Result - Verification	35
Figure 17: Accuracy Comparison	37
Figure 18: SVM Confusion Matrix.....	38
Figure 19: Random Forest Confusion Matrix.....	38
Figure 20: Gradient Boosting Confusion Matrix	39
Figure 21: X-Vector Confusion Matrix	39
Figure 22: ECAPA-TDNN Confusion Matrix.....	40

LIST OF TABLES

Table 1 : Summary of Related Works in Speaker Recognition	6
Table 2 : Software Libraries and Their Roles	12
Table 3 : Speaker Information of the Dataset	19
Table 4 : File Location (wav.scp)	25
Table 5 : Text Transcription	25
Table 6 : Utterance to Speaker Mapping (utt2spk)	25
Table 7 : Speaker Train / Test Summary	26
Table 8 : Comparison - ML vs Deep Learning Models	31
Table 9 : Model Identification Results.....	37
Table 10 : Speakers - Confusion Matrix Summary	40
Table 11 : Per-Speaker Average Confidence Score by Model (%)	41
Table 12 : Speaker Wise Summary of Precision of Different Models	42
Table 13 : Speaker Wise Summary of Recall of Different Models	42
Table 14 : Speaker Wise Summary of F1-Score of Different Models.....	43

LIST OF SYMBOLS AND ABBREVIATIONS

ML Machine Learning
DL Deep Learning
SVM Support Vector Machine
RF Random Forest
GB Gradient Boosting
MFCC Mel-Frequency Cepstral Coefficients
ECAPA-TDNN Emphasized Channel Attention, Propagation and Aggregation Time Delay
Neural Network
TDNN Time Delay Neural Network
WAV Waveform Audio File Format
GUI Graphical User Interface
ID Identification
EER Equal Error Rate
F1-Score Harmonic mean of Precision and Recall

CHAPTER 1:

INTRODUCTION

Voice is one of the most natural ways through which a person can be recognised. Unlike a password or PIN, a person does not need to remember it. Unlike an ID card or token, it does not need to be carried separately. Every person has a different way of speaking, which depends on the structure of the vocal tract, movement of the speech organs, pronunciation, accent, and speaking habits. Because of these differences, the voice can be used as a biometric feature for recognising or verifying a speaker.

Speaker recognition mainly answers two questions: **who is speaking** and **whether the speaker is really the person they claim to be**. The first task is called Speaker Identification. In this process, the system compares an input voice sample with the voices of registered speakers and predicts the most likely speaker. The second task is called Speaker Verification. In this process, the system checks whether a given voice sample matches a claimed identity and gives an output such as accepted or rejected.

Speaker identification and verification are useful in many practical applications. They can be used in voice-based login systems, telephone banking, forensic investigation, call-centre monitoring, attendance systems, and personalised digital assistants. As voice-based technologies are becoming more common, the need for reliable speaker recognition systems is also increasing.

In the early stages, speaker recognition systems mostly depended on manually extracted acoustic features such as Mel-Frequency Cepstral Coefficients, commonly known as MFCCs. These features were used with statistical models such as Gaussian Mixture Models and Hidden Markov Models. These methods worked well in controlled conditions, but their performance was affected in real-world situations. Background noise, different recording devices, emotional changes in speech, and limited training data often reduced the accuracy of such systems.

With the development of deep learning, speaker recognition has improved a lot. Modern deep learning models such as X-Vector and ECAPA-TDNN can learn speaker-specific patterns directly from speech data. These models can extract stronger speaker representations and are more suitable for handling variations in voice samples.

However, most speaker recognition research has been carried out for high-resource languages such as English and Mandarin. Many regional languages still have very limited support in this area. Assamese is one such language. It is spoken by more than 15 million people, mainly in Assam and nearby regions. Assamese is also widely used in education, media, administration, and daily communication. Even though it is an important language, very little work has been done on Assamese speech processing, especially in speaker identification and verification. There is also no widely available public benchmark dataset for Assamese speaker recognition.

This project tries to address this gap by developing a complete Speaker Identification and Speaker Verification system for Assamese speech. For this work, a custom Assamese speech dataset is

created using voice samples from 20 speakers. The dataset includes different voice characteristics across gender and age groups.

In this project, five models are implemented and evaluated. Three machine learning models, namely Support Vector Machine, Random Forest, and Gradient Boosting, are trained using MFCC-based features. Two deep learning models, X-Vector and ECAPA-TDNN, are used with log-Mel spectrogram-based inputs to extract stronger speaker features.

The system supports both speaker identification and speaker verification. In speaker identification, the system predicts the most likely speaker from a known set of speakers. In speaker verification, the system checks whether the voice sample belongs to the claimed speaker or not. The models are evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. Along with these metrics, confidence scores and similarity scores are also used during real-time prediction to show how sure the system is about its result.

Finally, a Streamlit-based web application is developed to demonstrate the system in real time. The application allows users to upload or record an audio sample and get speaker identification and verification results immediately. The complete workflow, from dataset creation to deployment, is designed to be simple, understandable, and reproducible. This project can also serve as a useful base for future research in Assamese speaker recognition.

CHAPTER 2:

LITERATURE REVIEW

Speaker recognition has been studied for many years, and the field has developed through different stages. Earlier systems mainly used statistical methods, later systems introduced machine learning classifiers, and recent systems mostly depend on deep learning-based speaker embeddings. This chapter reviews the important research work related to speaker identification and speaker verification and explains how the present project fits into this research area.

2.1 Classical and Statistical Approaches

Early speaker recognition systems were mainly based on statistical modelling techniques. Reynolds et al. [1] introduced the Gaussian Mixture Model–Universal Background Model approach, commonly known as GMM-UBM. This method became one of the most widely used techniques for text-independent speaker verification for many years. Kinnunen and Li [2] also provided an important survey on speaker recognition features and modelling methods. Their work helped establish Mel-Frequency Cepstral Coefficients, or MFCCs, as one of the most effective and commonly used feature extraction techniques in speaker recognition.

MFCC features became popular because they can represent the important spectral characteristics of speech, which are useful for distinguishing one speaker from another. Several studies have shown that MFCC-based systems can produce good results, especially when the recording conditions are controlled.

Welling and Kenn [S4] studied the use of MFCC features with Vector Quantisation for speaker modelling. Sriskandarajah and Sridharan [3] developed a real-time speaker identification system using MFCC and Vector Quantisation codebooks on the TIDIGITS corpus. Their work showed that even simple and compact feature representations can perform well in limited and controlled environments. Jayanna and Gurugopinath [4] compared standard MFCC and shifted MFCC features with fuzzy decision techniques and found that combining decision strategies can improve system performance under noisy conditions.

Some researchers also explored acoustic features other than MFCC. Ferras et al. [6] used Linear Predictive Coding and Discrete Wavelet Transform along with MFCC for Algerian Berber speech. Their results showed that combining different types of spectral and temporal features can improve recognition accuracy. Similarly, Kachouri et al. [8] combined pitch-based features with MFCC statistics to capture additional speaker-related information that may not be fully represented by MFCC alone. These studies show that using a richer set of acoustic features can help improve speaker recognition performance.

2.2 Machine Learning Classifiers

With the growth of machine learning, discriminative classifiers began to be used in speaker recognition systems. Support Vector Machine, or SVM, became popular because of its strong performance in classification tasks. Kwon and Narayanan [5] used multiclass SVM with MFCC-SDC features for emotional speaker recognition using the IEMOCAP corpus. Their study showed that SVM can capture speaker-related variations even when speech is affected by emotion or session differences.

Random Forest classifiers have also been used in speaker identification tasks. Since Random Forest is an ensemble-based method, it can reduce the effect of overfitting and handle variations in small datasets. This makes it useful for custom datasets where the number of speakers and samples may be limited. Gradient Boosting has also shown good performance in related audio classification tasks, especially when statistical features such as MFCC mean and standard deviation are used as input.

These machine learning methods are important because they are easier to train compared to deep learning models and can perform well when the dataset size is small. In the present project, SVM, Random Forest, and Gradient Boosting are used with MFCC-based features for Assamese speaker identification.

2.3 Deep Learning and Speaker Embeddings

Recent speaker recognition research has moved strongly towards deep learning. The introduction of speaker embeddings changed the way speaker recognition systems are designed. Snyder et al. [S1] introduced the x-vector approach, where a Time-Delay Neural Network is used to extract fixed-length speaker embeddings from speech signals. These embeddings represent speaker-specific information and can be used for both identification and verification tasks.

In x-vector-based systems, verification is usually performed by comparing speaker embeddings using cosine similarity or by using a backend classifier such as Probabilistic Linear Discriminant Analysis. This approach improved performance compared to older systems because the model learns speaker-discriminative patterns directly from data.

Desplanques et al. [S2] introduced ECAPA-TDNN, which is an improved version of the TDNN-based speaker recognition architecture. ECAPA-TDNN uses advanced techniques such as Res2Net blocks, feature aggregation, and attentive statistics pooling. These components help the model focus on the most useful parts of the speech signal for speaker recognition. Because of its strong performance, ECAPA-TDNN has become one of the most effective models for speaker verification.

Ravanelli et al. [3] introduced SpeechBrain, an open-source toolkit for speech processing. It provides pre-trained models and recipes for speaker recognition, making it easier to use advanced deep learning architectures. Deep learning models generally use log-Mel spectrograms or

filterbank features as input because they preserve more detailed spectral information than traditional MFCC features. These features allow neural networks to learn stronger speaker representations.

Although deep learning models require more data and computational resources, they have shown excellent results in speaker recognition. In the present project, X-Vector and ECAPA-TDNN are used to extract deeper speaker representations for Assamese speech.

2.4 Speaker Recognition for Indian and Assamese Languages

Compared to English and other high-resource languages, speaker recognition research for Indian languages is still limited. Assamese, despite being spoken by a large population, has received very little attention in speech and speaker recognition research.

Goswami and Kalita [23] developed an ANN-based Assamese speaker identification system using MFCC and LPC features on a self-recorded dataset of ten speakers. Their work is important because it is one of the early studies related to Assamese speaker identification. However, the system focused only on speaker identification and did not include speaker verification or deep learning models.

Phukan and Sarma [32] proposed a neural framework for Assamese speaker identification by considering dialectal variations. Their system used a Self-Organising Map with a Probabilistic Neural Network and worked with different regional varieties of Assamese speech. This study showed that dialect variation in Assamese language can make speaker recognition more challenging.

The present project builds on these earlier works by developing a more complete Assamese speaker recognition system. A custom dataset of 20 speakers is created, and five different models are implemented and compared. These include three machine learning models, namely SVM, Random Forest, and Gradient Boosting, and two deep learning models, X-Vector and ECAPA-TDNN. Unlike many previous works, this project supports both speaker identification and speaker verification. It also evaluates the models using accuracy, precision, recall, F1-score, confidence score, and similarity score.

2.5 Summary Table of Related Works

Table 1: Summary of Related Works in Speaker Recognition

Sl. No.	Title	Dataset Used	Technique Used	Validation Method	Ref.
1	Assamese Speaker Identification Using ANN	Self-collected Assamese speech, 10 speakers	MFCC and LPC features with an Artificial Neural Network classifier	Accuracy	[20]
2	Dialect-Sensitive Neural Framework for Assamese Speaker Identification	5 speakers across four Assamese regional dialects	SOM embedded in a Probabilistic Neural Network combined with Learning Vector Quantisation (LVQ)	Accuracy	[21]
3	MFCC-GMM Speaker Identification for Hindi Speech	15 speakers (10 male, 5 female)	MFCC features combined with a Gaussian Mixture Model classifier	Accuracy	[1]
4	SVM-Based Emotional Speaker Recognition with MFCC-SDC	IEMOCAP emotional speech database	MFCC augmented with Shifted Delta Cepstra fed into a multiclass SVM	Accuracy	[5]
5.	Speaker Recognition using Shifted MFCC and Fuzzy Logic	Speech Corpus	Shifted MFCC + VQ + Fuzzy Logic	Accuracy	[4]
6.	Distance-Based Similarity Measurement for Speaker Verification	FSCSR Corpus	Distance-based Scoring	EER	[27]
7.	Deep Speaker: End-to-End Neural Speaker Embedding System	Large-Scale Speech Corpus	Deep Speaker Embeddings	Accuracy	[25]
8.	Generalized End-to-End Loss for Speaker Verification	Speaker Verification Corpus	GE2E Deep Embeddings	EER	[26]

9.	X-Vectors: Robust DNN Embeddings for Speaker Recognition	VoxCeleb / NIST SRE	X-Vector Architecture	EER, minDCF	[9]
10.	ECAPA-TDNN for Speaker Verification	VoxCeleb1 & VoxCeleb2	ECAPA-TDNN	EER	[10]
11.	Real-Time Speaker Recognition using MFCC and VQ	TIDIGIT Database	MFCC + Vector Quantization	Accuracy	[3]
12.	Automatic Speaker Recognition using MFCC and ANN	IITG Speaker Recognition Database	MFCC + MLP Neural Network	Accuracy	[28]

The table above highlights two key gaps that the current project addresses: the near-complete absence of speaker recognition work for the Assamese language and the lack of a unified system that evaluates both machine learning and deep learning models with a verification component on the same custom dataset.

CHAPTER 3:

INITIAL SYSTEM STUDY

3.1 The Existing System

Most existing speaker recognition systems and research works are mainly focused on high-resource languages, especially English. Many systems are trained using large datasets such as VoxCeleb, and several cloud-based voice recognition APIs also perform well on standard benchmarks. However, these systems are not specifically developed for Assamese speech.

In many educational and research environments, speaker identification is still done manually by listening to voice samples or by maintaining separate voice records. This type of method is not very reliable because it depends on human judgement. It is also time-consuming and difficult to manage when the number of speakers increases.

Although some open-source speaker recognition toolkits are available, they often require good knowledge of deep learning frameworks and speech processing tools. For beginners or small research teams, these systems can be difficult to understand, modify, and apply to low-resource languages such as Assamese.

3.2 Drawbacks of the Existing System

The main limitations of the existing systems are:

Resource intensity: Most advanced speaker recognition systems require large labelled datasets and GPU-based training.

Framework complexity: Tools such as Kaldi and SpeechBrain are powerful, but they require specialised knowledge. This creates a steep learning curve for new users.

Language limitation: At present, there is no widely available public dataset, benchmark, or pre-trained model specifically designed for Assamese speaker recognition.

Limited flexibility: Many existing systems do not easily support small custom datasets. They also do not provide a simple way to compare different machine learning and deep learning models within the same framework.

Lack of verification capability: Several lightweight speaker recognition systems mainly focus on speaker identification only. They do not include speaker verification, which is necessary for authentication-based applications such as voice login.

3.3 Problem Definition

The main objective of this project is to design and implement an efficient speaker recognition system for the Assamese language using a custom speech dataset. The system aims to recognise speakers based on their voice characteristics and should work effectively on standard computing resources without depending on expensive hardware or very large datasets.

The proposed system focuses on two important tasks of speaker recognition:

Speaker Identification

Speaker Identification is the process of finding out who is speaking from a group of registered speakers. In this task, the system compares an input voice sample with all the stored speaker models and predicts the speaker whose voice features match the input sample most closely.

For example, suppose the database contains voice samples of Speaker 1, Speaker 2, Speaker 3, and Speaker 4. When an unknown voice sample is given to the system, it extracts the features from that voice and predicts which registered speaker is most likely to have spoken.

Speaker Verification

Speaker Verification is the process of checking whether a voice sample belongs to a claimed speaker. Unlike speaker identification, it does not search through all speakers. Instead, it verifies one specific identity claim.

In this case, the user first claims an identity, and the system checks whether the given voice matches the stored voice model of that claimed speaker.

Example:

Claimed Identity	Voice Sample	Result
Speaker 1	Voice of Speaker 1	Accepted
Speaker 2	Voice of Speaker 1	Rejected
Speaker 3	Voice of Speaker 3	Accepted

If the similarity score between the input voice and the stored voice model is higher than a predefined threshold, the claim is accepted. Otherwise, it is rejected.

Speaker verification is called **one-to-one matching** because the system compares the input voice only with the claimed speaker's voice model.

Project Objective

The proposed system combines both speaker identification and speaker verification capabilities within a single framework. Machine learning models such as Support Vector Machine (SVM), Random Forest (RF), and Gradient Boosting (GB) are employed for identification, while deep learning models such as X-Vector and ECAPA-TDNN are used to generate robust speaker embeddings for verification. The system also provides confidence and similarity scores to indicate the reliability of each prediction, making it suitable for real-time speaker recognition applications.

Difference Between Speaker Identification and Speaker Verification

Feature	Speaker Identification	Speaker Verification
Purpose	Determine who is speaking	Verify whether a claimed identity is correct
Matching Type	One-to-Many	One-to-One
Output	Predicted Speaker ID	Accept or Reject
Example Question	"Who is speaking?"	"Is this really Speaker A?"
Typical Applications	Attendance systems, Speaker indexing	Authentication, Voice-based login

3.4 The Proposed System

The proposed system is a modular, end-to-end speaker recognition pipeline consisting of the following components:

- **Data collection and preprocessing:** A custom Assamese speech dataset from 20 speakers, processed through silence removal, amplitude normalisation, and resampling.
- **Dual feature extraction:**
 - o MFCC with delta, delta-delta, and spectral features for ML models.
 - o Log-Mel spectrograms for deep learning models.
- **Model training:**
 - o Machine Learning: SVM, Random Forest, Gradient Boosting.
 - o Deep Learning: X-Vector and ECAPA-TDNN.

- **Verification strategies:**

- o Probability-based confidence scoring for ML models.

- o Cosine similarity-based scoring for DL models.

- **Real-time application:** A Streamlit-based web interface that enables audio upload and live recording for speaker identification and verification.

- **Evaluation and reporting:** Automated export of performance metrics, including accuracy, precision, recall, F1-score, confusion matrices, and per-sample confidence scores.

3.5 Scope of the System

The system implements a complete pipeline, beginning with raw audio input and ending with speaker recognition output. It supports:

- Closed-set identification, where all speakers are known during training.
- Binary verification, where a claimed identity is validated.

In addition to dataset-level evaluation, the system provides confidence and similarity scores during real-time inference, indicating the model's certainty for individual predictions.

The system is designed to run on a standard laptop without requiring a GPU, making it accessible for academic and small-scale research purposes.

3.6 Scope of this Project

The following tasks have been completed within the scope of this project:

- Design and recording of a 20-speaker Assamese speech dataset.
- Implementation of a preprocessing pipeline, including Praat-based annotation and automated silence removal.
- Feature extraction using MFCC and log-Mel spectrogram techniques.
- Training and evaluation of five models across machine learning and deep learning approaches.
- Implementation of confidence-based identification and verification mechanisms.
- Development of a Streamlit web application for real-time speaker recognition.
- Automated generation of evaluation reports, including classification metrics, confusion matrices, and confidence scores.

CHAPTER 4:

SOFTWARE AND HARDWARE REQUIREMENTS

4.1 Integrated Development Environment (IDE)

Visual Studio Code (VS Code) is used as the primary development environment, providing strong Python support, integrated terminal, debugging, Git integration, and productivity extensions.

4.2 Programming Language

Python 3.10+ is used throughout the project. Its rich ecosystem of speech processing, machine learning, and deep learning libraries makes it ideal for speaker recognition research.

4.3 Libraries and Frameworks

Table 2: Software Libraries and Their Roles

Library	Version	Purpose
librosa	0.10+	Audio loading, MFCC & mel-spectrogram extraction, silence trimming
NumPy	1.24+	Numerical operations, feature aggregation
scikit-learn	1.4+	SVM, Random Forest, Gradient Boosting, metrics, train/test split
PyTorch	2.1+	X-Vector and ECAPA-TDNN model definition and training
Streamlit	1.32+	Interactive web application for identification and verification
Matplotlib	3.8+	Accuracy plots and result visualisation
Seaborn	0.13+	Confusion matrix heatmaps
openpyxl	3.1+	Excel report generation (metrics, dataset splits)
SoundFile	0.12+	WAV file reading and writing
Praat	6.x	Speech annotation and TextGrid file generation

4.4 Operating System

The system is developed and tested on Windows 10/11, which provides stable Python development support, machine learning library compatibility, and seamless integration with external microphones for live recording.

4.5 Hardware Requirements

- Processor: Intel Core i3 (11th Generation) or higher.
- RAM: Minimum 8 GB (16 GB recommended for DL training).
- Storage: At least 10 GB free for audio data, features, and trained models.
- Microphone: Standard microphone for live recording in the Streamlit application.
- GPU: Not mandatory; however, the use of a CUDA-compatible GPU can significantly accelerate the training of deep learning models.

CHAPTER 5:

SYSTEM DESIGN

5.1 Work Flow Diagram

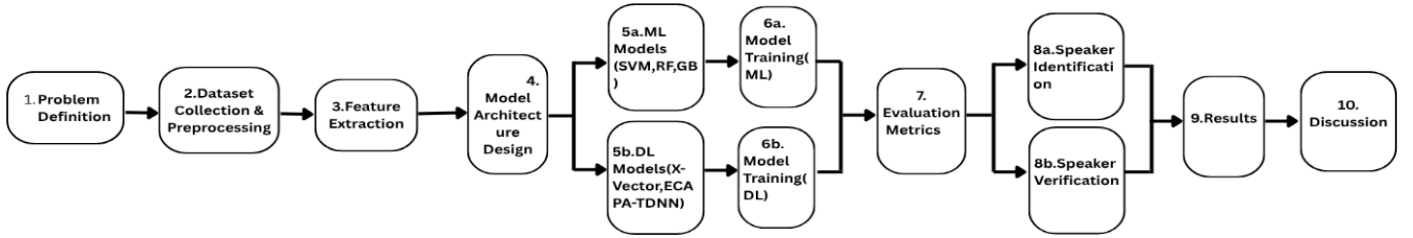


Figure 1: Work Flow Diagram

The proposed speaker recognition system follows a structured workflow that combines both machine learning and deep learning techniques for Assamese speaker identification and verification. The complete process begins with problem formulation and ends with performance analysis and result interpretation.

1. Problem Definition

The objective of this work is to develop a speaker recognition system capable of performing both speaker identification and speaker verification for Assamese speech. The system is designed to recognize speakers using their vocal characteristics and provide reliable authentication through voice-based analysis.

The two primary tasks addressed by the system are:

- **Speaker Identification:** Determining the identity of an unknown speaker from a set of enrolled speakers.
- **Speaker Verification:** Confirming whether a speech sample belongs to a claimed speaker.

To achieve these objectives, both traditional machine learning models and modern deep learning architectures are incorporated within a unified framework.

2. Dataset Collection and Preprocessing

A custom Assamese speech dataset is created by collecting speech recordings from multiple speakers. All recordings are stored in WAV format and processed to ensure uniformity and quality before feature extraction.

The preprocessing stage includes:

- Removal of silent portions from recordings using a threshold-based trimming technique.
- Amplitude normalization to maintain consistent signal levels.
- Resampling of audio signals to a sampling rate of 16 kHz.
- Speech segmentation and annotation using Praat.

These operations reduce unwanted variations in the recordings and improve the reliability of the extracted speech features.

3. Feature Extraction

The feature extraction stage transforms raw speech signals into compact numerical representations that can be utilized by machine learning and deep learning models.

3.1 MFCC Feature Extraction for Machine Learning

For machine learning-based recognition, Mel Frequency Cepstral Coefficients (MFCCs) are extracted from the speech recordings. In addition to static MFCCs, dynamic speech characteristics are captured through delta and delta-delta coefficients. Various statistical measures are then computed to generate fixed-length feature vectors suitable for classification.

The extracted features effectively represent speaker-specific vocal characteristics while reducing the dimensionality of the original speech signal.

3.2 Log-Mel Spectrogram Extraction for Deep Learning

For deep learning models, speech recordings are converted into normalized log-Mel spectrograms. These spectrograms preserve both temporal and spectral information that is important for learning robust speaker representations.

To maintain a consistent input size, each spectrogram is padded or truncated to a fixed length before being supplied to the deep learning architectures.

4. Model Architecture Design

The proposed framework consists of two parallel modelling branches, allowing a comparative analysis between machine learning and deep learning approaches.

4.1 Machine Learning Architecture

The machine learning branch utilizes feature vectors derived from MFCC-based representations. Before classification, the features are standardized to improve learning performance.

The classifiers employed in this branch include:

- Support Vector Machine (SVM)
- Random Forest (RF)
- Gradient Boosting (GB)

These algorithms are selected because of their effectiveness in handling speaker classification tasks on relatively small datasets.

4.2 Deep Learning Architecture

The deep learning branch employs advanced speaker embedding models capable of learning discriminative speaker characteristics directly from speech data.

The architectures used are:

- X-Vector
- ECAPA-TDNN

These models generate compact speaker embeddings that capture unique voice characteristics and can be utilized for both identification and verification.

5. Model Implementation

5.1 Machine Learning Models

The MFCC-based feature vectors are provided to the machine learning classifiers for training and prediction. During inference, these models generate class probabilities that can be interpreted as confidence scores for speaker identification.

5.2 Deep Learning Models

The deep learning branch processes speech signals through pretrained speaker embedding networks. The resulting embeddings are stored and compared using similarity-based techniques to determine speaker identity and authenticity.

6. Model Training

6.1 Training of Machine Learning Models

The machine learning models are trained using labelled speech samples collected from enrolled speakers. The dataset is divided into training and testing subsets to evaluate the generalization capability of the models. Appropriate parameter tuning is performed to achieve optimal classification performance.

6.2 Speaker Embedding Generation

The deep learning models are used to generate speaker embeddings from speech recordings. These embeddings serve as compact representations of speaker characteristics and are subsequently employed during identification and verification.

7. Performance Evaluation

The effectiveness of the proposed system is assessed using several quantitative evaluation metrics.

7.1 Identification Metrics

Speaker identification performance is measured using:

- Accuracy
- Precision
- Recall
- F1-Score
- Support

These metrics provide insight into the classification capability of the system for each enrolled speaker.

7.2 Verification Metrics

Speaker verification performance is evaluated by analysing:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

These measures indicate the system's ability to correctly accept genuine users and reject impostors.

7.3 Confidence-Based Analysis

To support real-time decision making, confidence scores are generated during prediction.

- Machine learning models provide probability-based confidence values.
- Deep learning models generate cosine similarity scores between speaker embeddings.

These values help determine the certainty of individual predictions and improve the interpretability of the system.

8. Speaker Recognition Tasks

8.1 Speaker Identification

In the identification stage, the system receives an unknown speech sample and predicts the most likely speaker from the enrolled speaker database. The prediction is based on the learned speaker characteristics extracted during training.

8.2 Speaker Verification

In the verification stage, a speaker claims a specific identity. The system compares the input speech sample with the stored voice representation of the claimed speaker and either accepts or rejects the claim based on a predefined similarity threshold.

9. Results

The performance of all implemented models, namely SVM, Random Forest, Gradient Boosting, X-Vector, and ECAPA-TDNN, is analysed and compared. Evaluation reports, confusion matrices, confidence scores, and verification results are automatically generated and stored for further analysis.

10. Discussion

The final stage focuses on interpreting the experimental results and assessing the overall effectiveness of the proposed system. Particular attention is given to:

- Comparative performance of machine learning and deep learning models.
- Influence of feature extraction techniques on recognition accuracy.
- Challenges associated with Assamese speech processing.
- Limitations of the current implementation.
- Opportunities for future enhancement, including larger datasets, improved verification strategies, and real-time deployment.

The proposed framework demonstrates that reliable speaker identification and verification can be achieved for Assamese speech using a combination of traditional machine learning methods and modern deep learning-based speaker embeddings, while remaining computationally feasible on standard computing hardware.

CHAPTER 6:

IMPLEMENTATION

6.1 Speech Dataset Creation

The speech dataset used in this study was self-collected from 20 Assamese speakers. All recordings were stored in WAV format with a sampling rate of 16 kHz and mono channel configuration. The dataset consists of 2,993 audio files, which were divided into training and testing sets using an 80:20 ratio.

Dataset Statistics:

- Number of Speakers: 20
- Total Audio Files: 2,993
- Training Files: 2,394
- Testing Files: 599
- Sampling Rate: 16 kHz
- Audio Format: WAV
- Total Duration: 1.40 hours

Table 3: Speaker Information of the Dataset

Sl. No	Speaker	Gender	Age	Profession	Address
1	Speaker 1	Male	23	Student	Golaghat
2	Speaker 2	Male	25	Student	Sualkuchi
3	Speaker 3	Male	19	Student	Guwahati
4	Speaker 4	Male	23	Student	Tezpur
5	Speaker 5	Male	15	Student	Nagaon
6	Speaker 6	Female	22	Student	Morigaon
7	Speaker 7	Male	23	Student	Charaideo
8	Speaker 8	Male	22	Student	Golaghat
9	Speaker 9	Male	24	Student	Charaideo
10	Speaker 10	Male	24	Student	Dhemaji
11	Speaker 11	Male	20	Student	Tezpur
12	Speaker 12	Male	25	Student	Jagiroad
13	Speaker 13	Male	23	Student	Nagaon

14	Speaker 14	Female	18	Student	Charaideo
15	Speaker 15	Female	26	Research Scholar	Guwahati
16	Speaker 16	Male	27	Research Scholar	Dibrugarh
17	Speaker 17	Female	27	Research Scholar	Nagaon
18	Speaker 18	Female	26	Research Scholar	Guwahati
19	Speaker 19	Male	28	Research Scholar	Tezpur
20	Speaker 20	Female	23	Student	Nagaon

Recording Environment and Conditions

All recordings were conducted in a relatively quiet indoor environment to minimize background noise and external disturbances. The same microphone and recording setup were used for all speakers to maintain uniform recording conditions and reduce channel variability.

Recording Protocol and Naming Convention

Each speaker was instructed to record a **single predefined sentence**, which was spoken **three times consecutively in one recording session**. This approach ensures consistency in phonetic content while allowing natural intra-speaker variations in voice, pitch, and speaking style.

Files are named:

ASS_S1_SEN1(1).wav

S1 = Speaker ID, SEN1 = Sentence ID, (1) = Recording instance

All recordings are in WAV format, 16 kHz, mono, 16-bit PCM.

6.2 Speech Annotation Using Praat

Praat was used to annotate Assamese audio recordings with TextGrid files, marking precise time boundaries for speech and silence segments. This enables accurate automated trimming.

Step 1 — Open Audio File

Open Praat → Praat Objects window → Open → Read from file → Select WAV file.

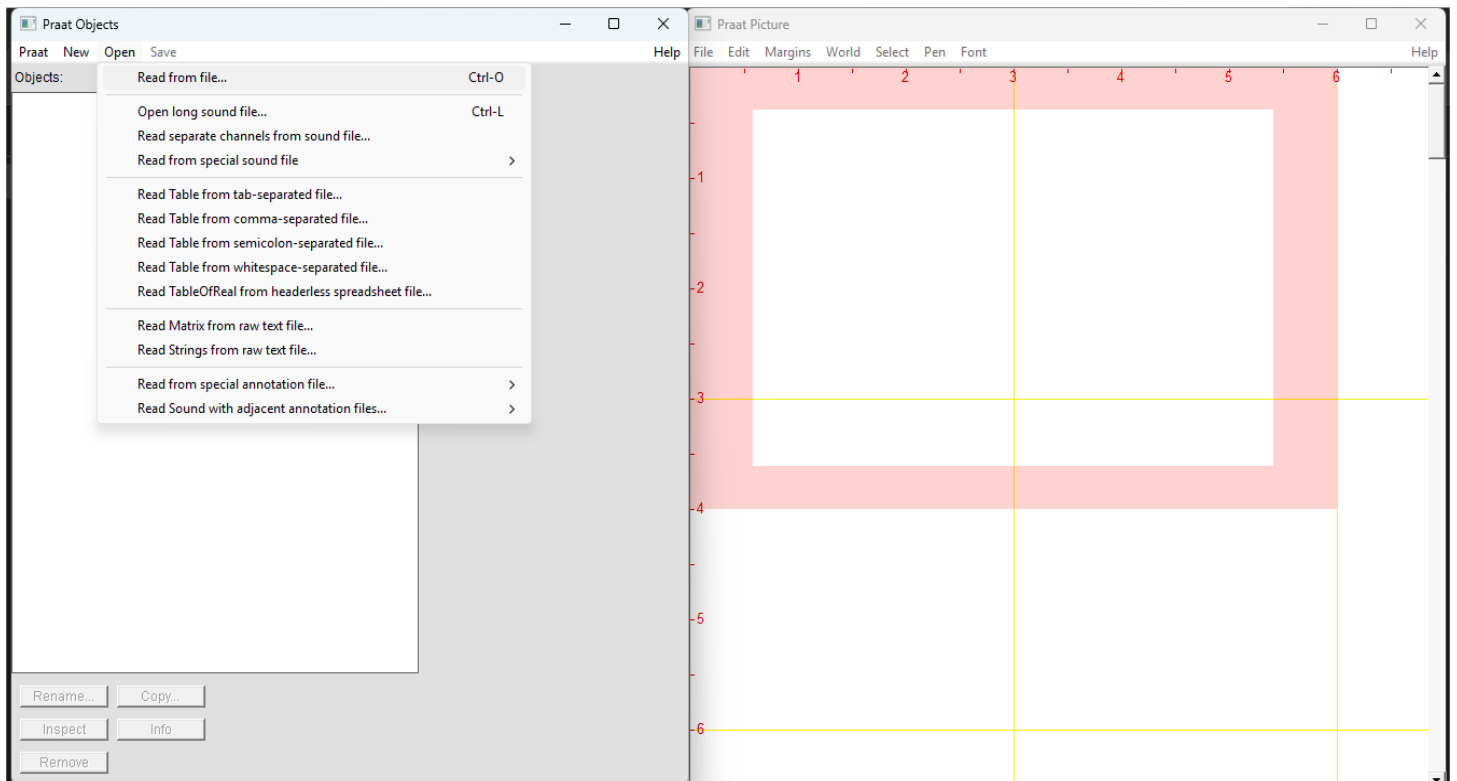


Figure 2: Praat Annotation - Read from File

Step 2 — Create Text Grid

Select audio object → Annotate → To TextGrid → Specify tier names (Words, Sentence).

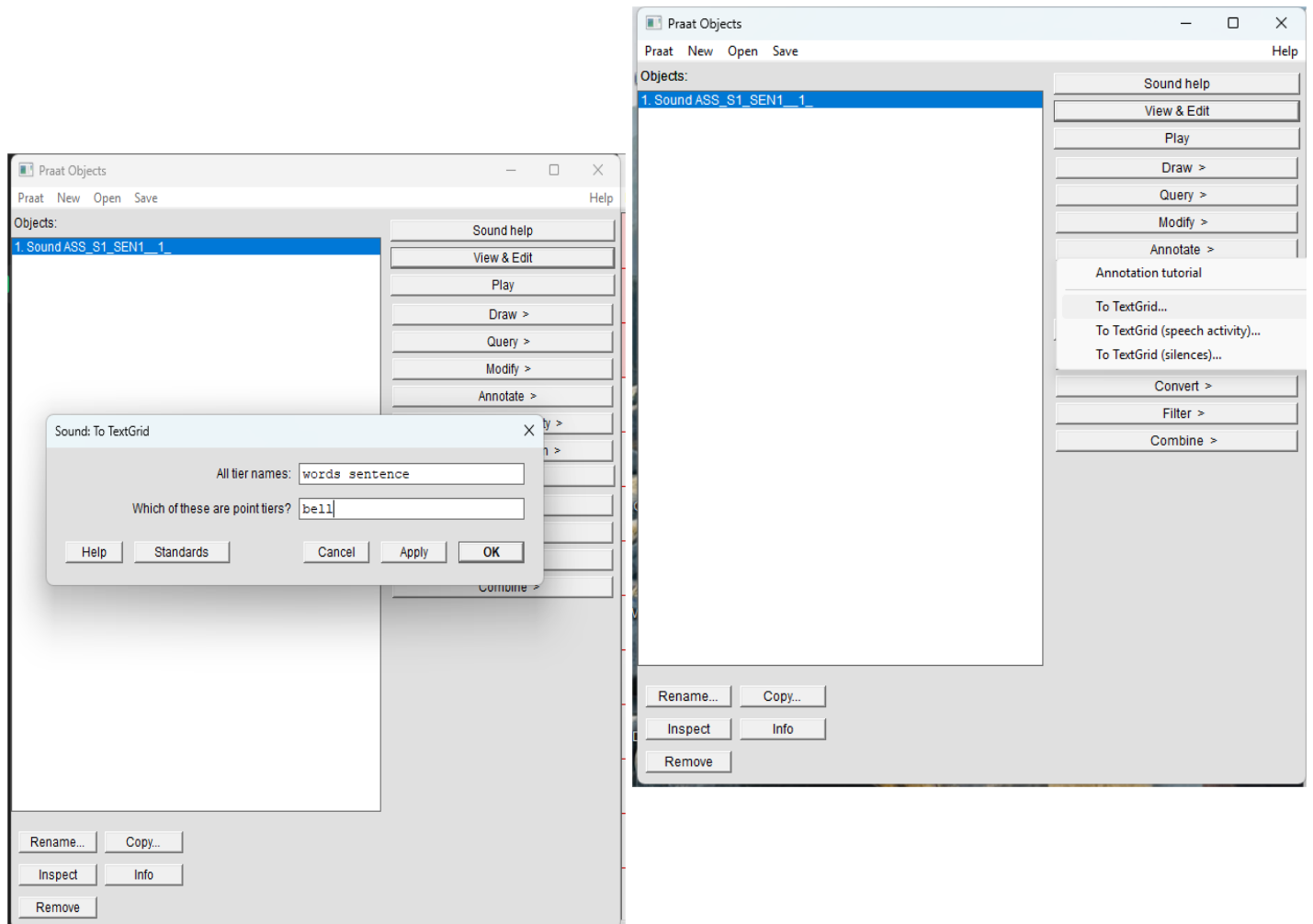


Figure 3: Praat Annotation - TextGrid Dialog

Step 3 — Add Annotation Tiers

Tiers added: id, Word id, English word, Words, English sentences, Sentences. These allow detailed linguistic annotation across multiple levels.

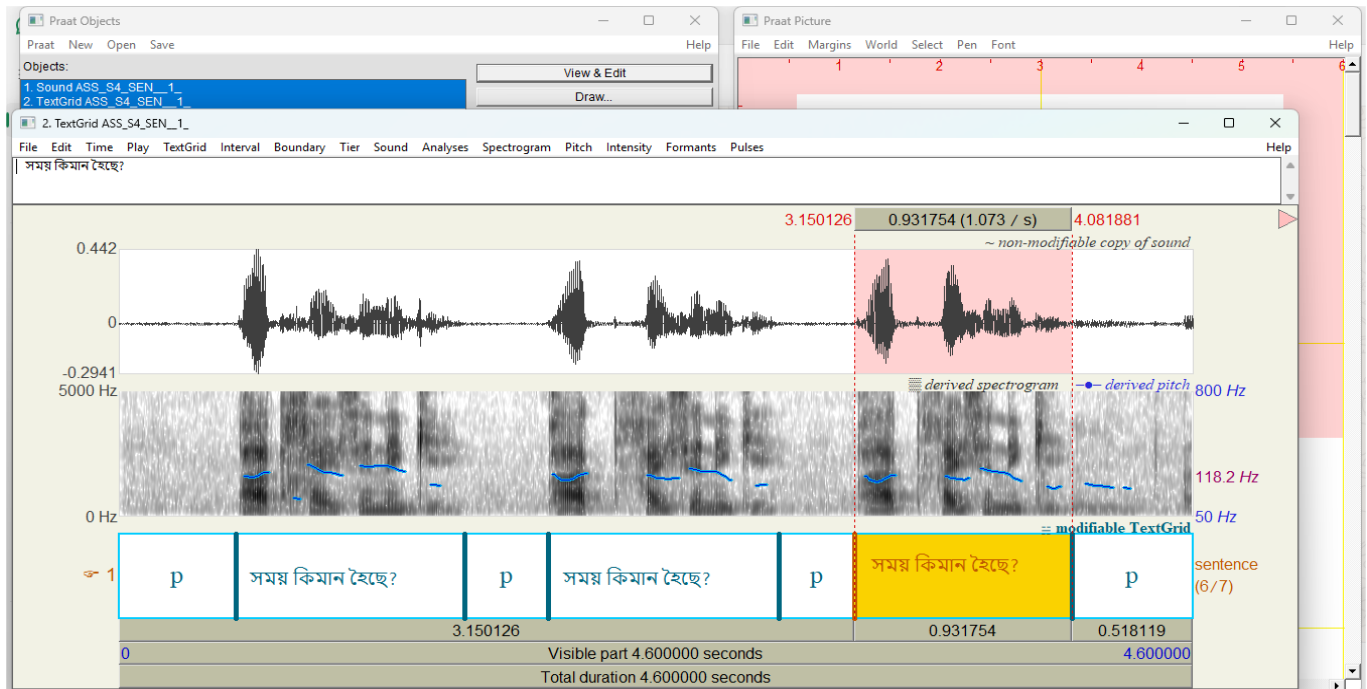


Figure 4: Praat Annotation - Tier Structure

6.3 Automated Audio Trimming Using Python

To ensure high-quality and consistent speech data, an automated audio trimming process was implemented using Python in combination with Praat TextGrid files.

Use of TextGrid Files for Segmentation

Praat **TextGrid** files are used to annotate speech recordings by marking precise time intervals corresponding to speech and silence. These annotations provide accurate segmentation boundaries required for trimming audio signals.

Python-Based Parsing of TextGrid Annotations

A Python script parses the TextGrid files to read the start and end timestamps of speech segments. The script identifies labeled regions containing valid speech while ignoring silence and non-speech intervals.

Automatic Extraction of Speech Segments

Using the extracted timestamps, the corresponding portions of the original WAV files are automatically segmented. This process ensures that only meaningful speech data is retained for further processing.

Generation of Trimmed WAV Files The extracted speech segments are saved as **new trimmed WAV files** with consistent sampling parameters (16 kHz, mono channel). The trimmed files are stored in a structured, speaker-wise directory format for efficient access during training and evaluation.

Verification of Trimmed Audio Quality

The quality of the trimmed audio samples is verified by:

- Auditory inspection through playback
- Ensuring absence of leading and trailing silence
- Confirming correct duration and speech clarity

This automated trimming approach improves dataset consistency, reduces manual effort, and enhances the overall performance of speaker identification and verification models.

6.4 Audio Preprocessing

Audio Data

We created a centralized speech dataset for our Speaker Identification and Verification system. The dataset contains audio recordings from 20 speakers, including 14 male and 6 female speakers. Each speaker recorded several Assamese speech samples so that differences in voice, tone, and speaking style could be captured.

All recordings are saved in WAV format under similar recording conditions to maintain consistency. The audio files are named in a simple and organized way, such as **ASS_S1_SEN1.wav** and **ASS_S2_SEN2.wav**, where the file name shows the speaker number and the sentence number. This naming method makes the dataset easy to manage and label.

Having a centralized and well-organized dataset helps improve the training and testing of the speaker identification and verification models

Acoustic Data

Acoustic data preparation involves organizing the audio files according to speaker identity and extracting important speech features. Features such as MFCCs and deep speaker embeddings are generated from the audio recordings. These features are then used to train and test machine learning and deep learning models for speaker identification and verification.

Table 4: File Location (wav.scp)

Spk ID	WAV File Location
S1	C:\Users\KRISHNAMONI\Downloads\Speaker_Identification\Dataset/SPEAKER 1/ASS_S1_SEN1(1).wav
S2	C:\Users\KRISHNAMONI\Downloads\Speaker_Identification\Dataset/SPEAKER 2/ASS_S2_SEN1(1).wav

Table 5: Text Transcription

Utterance ID	Transcription (Assamese)
ASS_S1_SEN4	সময় কিমান হৈছে?
ASS_S1_SEN5	ভাৰতবৰ্ষ আমাৰ দেশ।

Table 6: Utterance to Speaker Mapping (utt2spk)

Utterance ID	Speaker ID
ASS_S1_SEN4	S1
ASS_S1_SEN5	S1

6.5 Feature Extraction

MFCC Features (ML Models)

For ML models the following features are extracted and aggregated as mean + standard deviation over all time frames:

- 40 Mel-Frequency Cepstral Coefficients (MFCCs).

- First-order delta (velocity) and second-order delta-delta (acceleration) coefficients.
- Spectral Centroid, Spectral Rolloff, Spectral Contrast (7 bands).
- Zero-Crossing Rate (ZCR) and Root-Mean-Square Energy (RMS).

Log-Mel Spectrogram (DL Models)

An 80-band log-Mel spectrogram is computed with a 25 ms Hanning window and 10 ms hop. The spectrogram is normalised (zero mean, unit variance) and padded or truncated to 300 frames, giving a fixed input of shape $[1 \times 80 \times 300]$.

6.6 Model Training

Table 7: Speaker Train / Test Summary

Per-Speaker Train / Test Summary					
Total files: 2993 Train: 2394 Test: 599 Split ratio: 80/20 Speakers: 20					
Speaker Label	Original Folder	Total Files	Train Files	Test Files	Train %
Speaker_01	Speaker 1	93	74	19	79.6%
Speaker_02	Speaker 2	108	86	22	79.6%
Speaker_03	Speaker 3	177	141	36	79.7%
Speaker_04	Speaker 4	187	149	38	79.7%
Speaker_05	Speaker 5	87	70	17	80.5%
Speaker_06	Speaker 6	106	85	21	80.2%
Speaker_07	Speaker 7	177	142	35	80.2%
Speaker_08	Speaker 8	184	147	37	79.9%
Speaker_09	Speaker 9	150	120	30	80.0%
Speaker_10	Speaker 10	135	108	27	80.0%

Speaker_11	Speaker 11	195	156	39	80.0%
Speaker_12	Speaker 12	150	120	30	80.0%
Speaker_13	Speaker 13	132	106	26	80.3%
Speaker_14	Speaker 14	165	132	33	80.0%
Speaker_15	Speaker 15	180	144	36	80.0%
Speaker_16	Speaker 16	216	173	43	80.1%
Speaker_17	Speaker 17	185	148	37	80.0%
Speaker_18	Speaker 18	126	101	25	80.2%
Speaker_19	Speaker 19	150	120	30	80.0%
Speaker_20	Speaker 20	90	72	18	80.0%
TOTAL	20 speakers	2993	2394	599	80.0%

All training is performed via terminal using `train.py`. An 80/20 stratified train/test split (`random_state=42`) ensures equal speaker representation in both sets.

Machine Learning (ML) Training

The machine learning models are implemented using the Scikit-learn Pipeline framework, which ensures a structured and reproducible workflow by chaining preprocessing and classification steps. Each pipeline consists of a *StandardScaler* for feature normalization followed by a chosen classifier (such as Support Vector Machine, Random Forest, or Logistic Regression). Feature scaling is a critical step as it standardizes the input feature distribution, thereby improving convergence and overall model performance.

The dataset is divided into training and validation subsets to evaluate model generalization. During training, the pipeline automatically applies normalization before feeding the data into the classifier, eliminating the risk of data leakage and ensuring consistency during both training and inference.

phases. Hyperparameters for the classifiers may be tuned using cross-validation techniques to achieve optimal performance.

Once trained, the models are serialized using appropriate persistence methods (e.g., joblib or pickle) and stored in the *models/* directory. This allows efficient reuse of trained models without the need for retraining. The training process can be executed using the following command:
python train.py --mode ml

Deep Learning (DL) Training

The deep learning approach utilizes advanced speaker embedding architectures, specifically X-Vector and ECAPA-TDNN, which are well-suited for speaker recognition tasks due to their ability to capture high-level discriminative features from speech signals. These models are trained using the Adam optimizer with an initial learning rate of 0.001, providing adaptive gradient updates for faster convergence.

To further enhance training efficiency, a cosine annealing learning rate scheduler is employed, which gradually reduces the learning rate following a cosine decay pattern. This helps the model escape local minima and achieve better generalization. The training objective is defined using cross-entropy loss, which measures the discrepancy between predicted and true speaker labels.

Additionally, gradient clipping is applied during backpropagation to prevent exploding gradients and ensure numerical stability, particularly in deep architectures. The models are trained for a total of 30 epochs, with performance monitored periodically using validation data.

After the completion of training, a *speaker gallery* is automatically constructed. This involves computing the mean embedding vector for each speaker by averaging embeddings extracted from multiple speech samples. These representative embeddings are stored in the *pretrained_models/* directory and are later used for speaker identification and verification tasks.

The deep learning training process can be initiated using the following command:
python train.py --mode dl

6.7 Model Evaluation and Performance Analysis

Performance is evaluated using the following metrics:

Accuracy:

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted samples to the total number of samples.

The mathematical formula for Accuracy is

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Figure 5: Accuracy Formula

Where:

TP (True Positive): Correctly identified

TN (True Negative): Correctly rejected

FP (False Positive): Incorrectly accepted

FN (False Negative): Incorrectly rejected

Precision:

Fraction of positive predictions that are correct.

Precision measures how many of the predicted positive results are actually correct.

The mathematical formula for Precision is:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Figure 6: Precision Formula

Recall:

Recall, also called Sensitivity, measures how many actual positive samples are correctly identified by the model.

The mathematical formula for Recall is

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Figure 7: Recall Formula

F1-Score:

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced evaluation when both false positives and false negatives are important.

The mathematical formula for F1-Score is:

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 8: F1-Score Formula

Support:

Support represents the total number of actual samples belonging to a particular class or speaker in the dataset.

The mathematical formula for Support is:

$$\text{Support} = \text{TP} + \text{FN}$$

Figure 9: Support Formula

All results are automatically exported to:

- results/model_metrics.xlsx — Dashboard, Speaker Confidence, Model Report, PRF Graphs, Heatmap Grid.
- results/dataset_splits.xlsx — Full train/test split file list per speaker.
- results/cm_*.png — Confusion matrix images per model.

6.8 Models Used and Justification

Support Vector Machine (SVM)

SVM with an RBF kernel ($C=10$, $\gamma=\text{'scale'}$) provides robust decision boundaries in high-dimensional MFCC feature spaces. Platt scaling ($\text{probability}=\text{True}$) enables probability-based verification. It serves as a reliable traditional baseline.

Random Forest

An ensemble of 300 decision trees trained with bootstrap sampling improves accuracy and reduces overfitting. Class probabilities are the fraction of trees voting for each class, suitable for verification thresholding.

Gradient Boosting

200 sequential trees ($\text{learning rate}=0.1$, $\text{max depth}=5$) each correct errors of the previous ensemble, capturing complex feature interactions and often outperforming Random Forest on small datasets.

X-Vector (TDNN)

Five TDNN layers with dilated convolutions capture multi-scale temporal context from log-Mel spectrograms. Statistics pooling (mean + std) produces a 192-dimensional speaker embedding. It is the established deep learning baseline for speaker recognition.

ECAPA-TDNN

Improves on X-Vector with Res2Net residual blocks for multi-granularity feature learning, multi-scale feature concatenation across three blocks, and attentive statistics pooling that weights time frames by their speaker-discriminative importance. Consistently achieves state-of-the-art results.

Table 8: Comparison - ML vs Deep Learning Models

Aspect	SVM/RF/GB	X-Vector	ECAPA-TDNN
Feature Input	MFCC + Spectral	Log-Mel Spectrogram	Log-Mel Spectrogram
Verification	Probability-based	Cosine similarity	Cosine similarity
Performance	Moderate–Good	Good	Best
Noise Robustness	Sensitive	Robust	Most robust
Interpretability	High	Low	Low

6.9 Web-Based Application Development

The Streamlit application provides a browser-based interface for real-time identification and verification without any coding required from the user.

Interface Design

A single-page layout presents a Model Type dropdown (Machine Learning / Deep Learning), audio input options, and separate identification and verification sections with results.

Speaker Identification & Verification System

Select Model Type

Machine Learning

Audio Input Method

☒ Upload WAV

☐ Record Audio

Upload WAV file

Drag and drop file here
Limit 200MB per file • WAV

Browse files

• **Machine Learning Speaker Identification**

Select ML Model

SVM

Figure 10: Streamlit Web Interface

Model Selection

Users select from ML (SVM, Random Forest, Gradient Boosting) or DL (X-Vector, ECAPA-TDNN). The interface dynamically updates controls for the chosen paradigm.

Speaker Identification & Verification System

Select Model Type

Machine Learning

Machine Learning

Deep Learning

Audio Input Method

☒ Upload WAV

☐ Record Audio

Upload WAV file

Drag and drop file here
Limit 200MB per file • WAV

Browse files

Deploy

Model Status

SVM

Random Forest

Gradient Boosting

X-Vector

ECAPA-TDNN

Speakers trained: 20

> Speaker list

Device: CPU

Figure 11: Model Selection - Machine Learning

Speaker Identification & Verification System

Select Model Type

Deep Learning

Audio Input Method

☒ Upload WAV
☐ Record Audio

Upload WAV file

Drag and drop file here
Limit 200MB per file • WAV

Browse files

• **Deep Learning Speaker Identification**

Select DL Model

X-Vector

X-Vector

ECAPA-TDNN

Figure 12: Model Selection - Deep Learning

Audio Input

Users upload a WAV file or record directly via the browser microphone. A waveform is displayed after loading. Both paths use the same preprocessing pipeline.

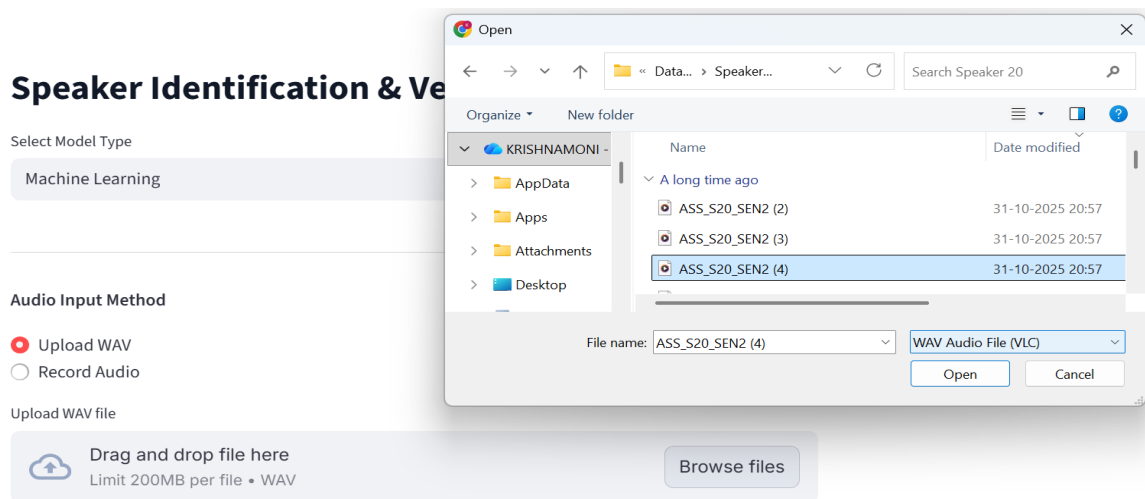


Figure 13: Select Voice

Speaker Identification & Verification System

Select Model Type

Machine Learning

Audio Input Method

- ☒ Upload WAV
☐ Record Audio

Upload WAV file



Drag and drop file here
Limit 200MB per file • WAV

Browse files



ASS_S13_SEN40(3).wav 165.2KB



0:01 / 0:01



Figure 14: Voice loaded

Display Results

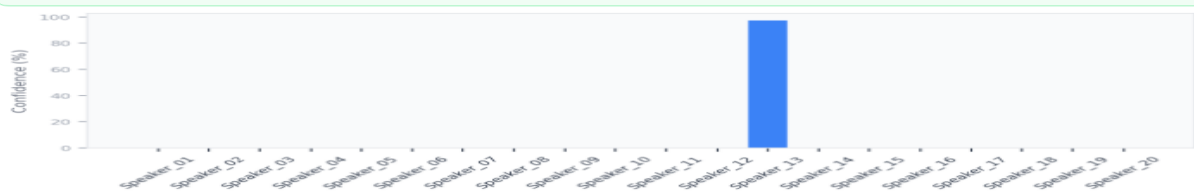
Identification shows the predicted speaker with confidence percentage and a probability bar chart. Verification shows ACCEPT or REJECT with the underlying score and an adjustable threshold slider.

Machine Learning Speaker Identification

Select ML Model

SVM

Identified Speaker: **Speaker_13** 97.8%



Machine Learning Speaker Verification

Claimed Speaker

Speaker_01

Acceptance Threshold (probability)

0.50

Figure 15: Predicted Result - Identification

♦ Machine Learning Speaker Verification

Claimed Speaker

Speaker_13

Acceptance Threshold (probability)

0.50

Verify using model

SVM

ACCEPT — Speaker_13 score 0.978

▼ All ML models verdict

SVM	ACCEPT score=0.978
Random Forest	ACCEPT score=0.840
Gradient Boosting	ACCEPT score=1.000

Figure 16: Predicted Result - Verification

6.10 System Integration and Testing

The system was developed using a modular architecture, and its components were integrated to form a complete end-to-end speaker recognition pipeline. The overall workflow connects multiple stages, beginning with audio input acquisition and proceeding through preprocessing, feature extraction, model inference, and finally result visualization through a user-friendly Streamlit interface.

In the first stage, the system accepts audio input either through file upload or recording. This input is then passed to the preprocessing module, where operations such as noise reduction, silence trimming, and normalization are performed to enhance audio quality and ensure consistency. The preprocessed signal is subsequently fed into the feature extraction module, where relevant acoustic features (e.g., MFCCs or learned embeddings in the case of deep learning models) are generated. These extracted features are then provided to the trained models for inference. Depending on the selected mode, the system performs either speaker identification (determining the most likely speaker from a known set) or speaker verification (confirming whether the input matches a claimed identity). The inference results are processed and displayed to the user through the Streamlit interface, ensuring clarity and ease of interpretation.

To evaluate the robustness and reliability of the integrated system, extensive functional testing was conducted using unseen audio samples that were not part of the training dataset. This ensures that the system's performance reflects real-world conditions and generalization capability. The testing process included the following key aspects:

- **Correct Identification:** Verifying that the system accurately identifies speakers from the enrolled dataset.
- **Correct Verification:** Ensuring reliable acceptance or rejection decisions in speaker verification tasks.
- **Robustness to Invalid Inputs:** Assessing the system's ability to handle edge cases such as non-audio files, extremely short audio clips, and recordings with significant background noise.

The system demonstrated stable performance across these scenarios, successfully handling erroneous inputs without crashes and providing appropriate feedback to the user. This confirms the effectiveness of the integration process and the readiness of the system for practical deployment.

CHAPTER 7:

RESULTS

This chapter presents experimental results for speaker identification and verification across all five models, including accuracy, precision, recall, F1-score, speaker-wise confidence, and 2×2 confusion matrices.

7.1 Identification Performance

Table 9: Model Identification Results

Model	Type	Accuracy	Macro F1	Weighted F1
SVM	ML	[99.2%]	[99.1%]	[99.2%]
Random Forest	ML	[99.7%]	[99.7%]	[99.7%]
Gradient Boosting	ML	[96.2%]	[95.9%]	[96.1%]
X-Vector	DL	[99.3%]	[99.3%]	[99.3%]
ECAPA-TDNN ★	DL	[99.7%]	[99.7%]	[99.7%]

★ Best performing model. [99.7%]

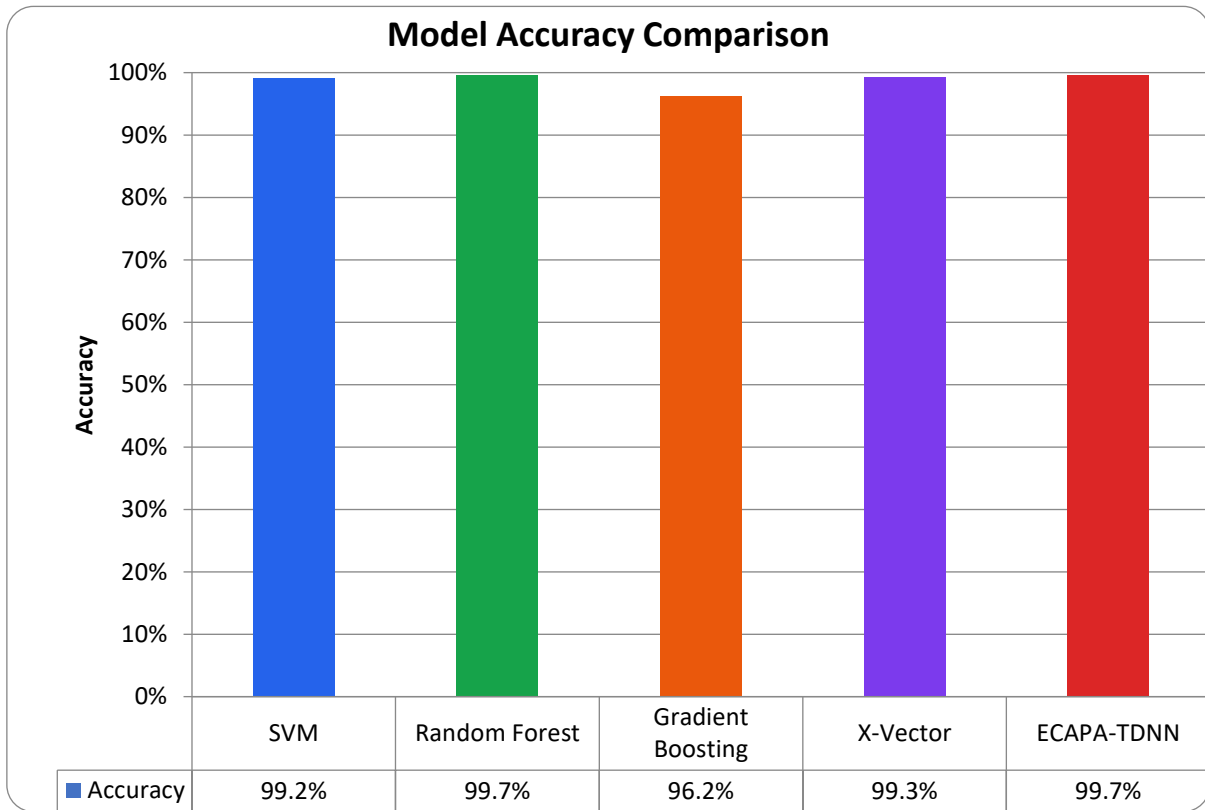


Figure 17: Accuracy Comparison

7.2 Speaker 2×2 Confusion Matrix

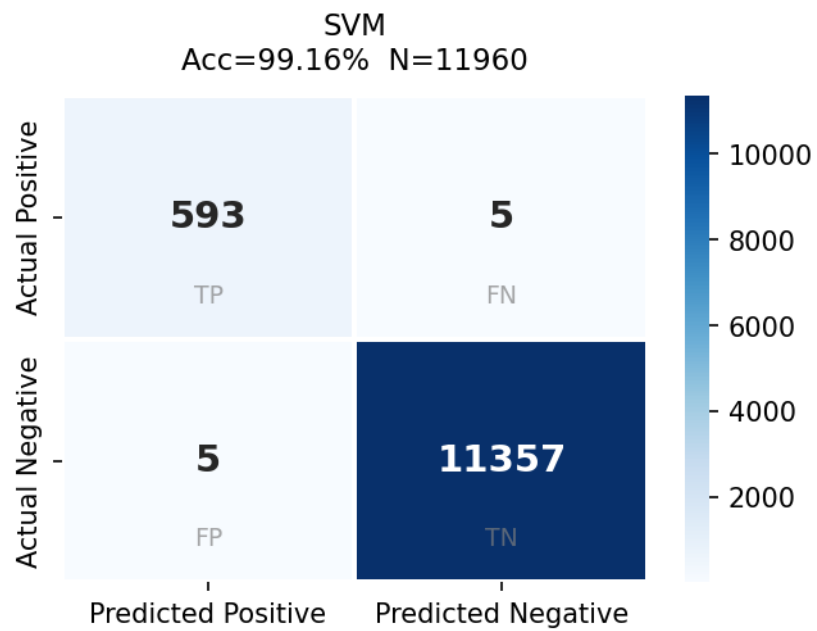


Figure 18: SVM Confusion Matrix

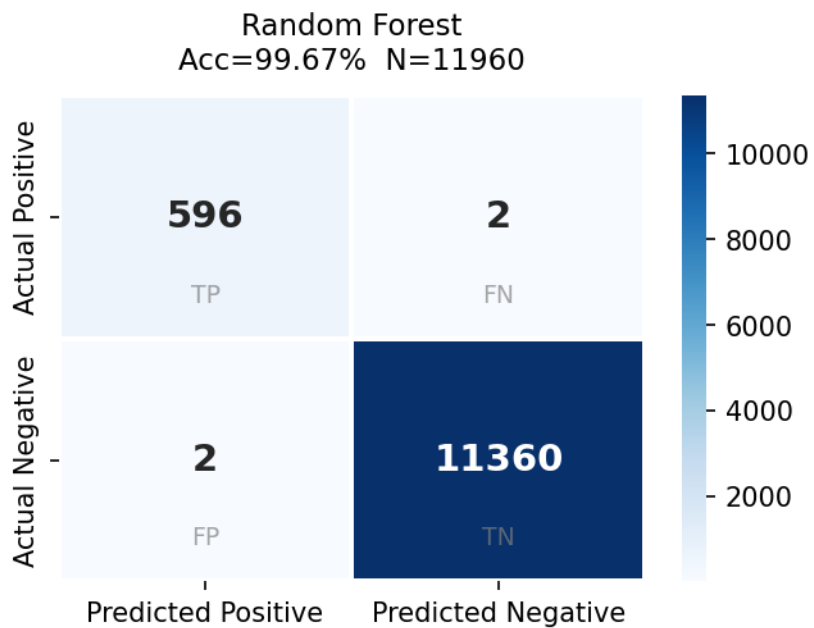


Figure 19: Random Forest Confusion Matrix

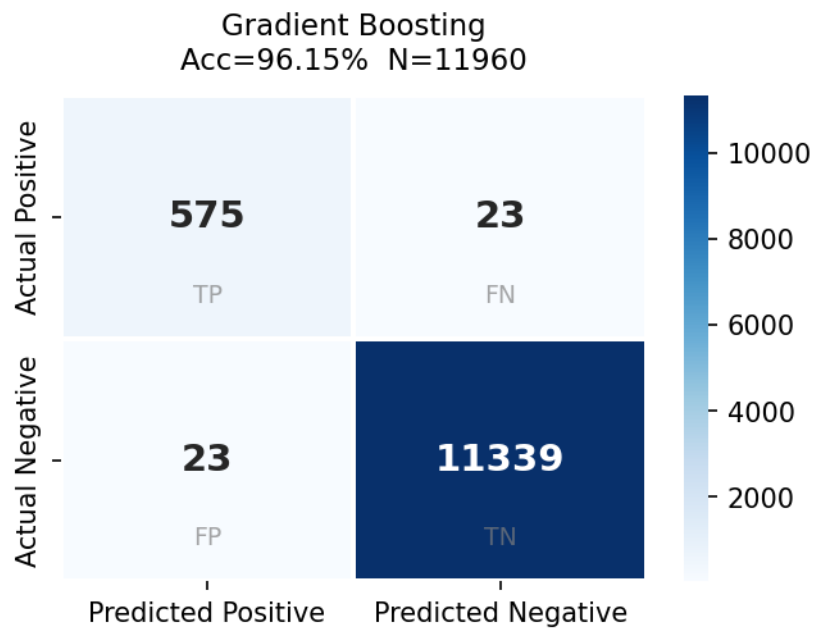


Figure 20: Gradient Boosting Confusion Matrix

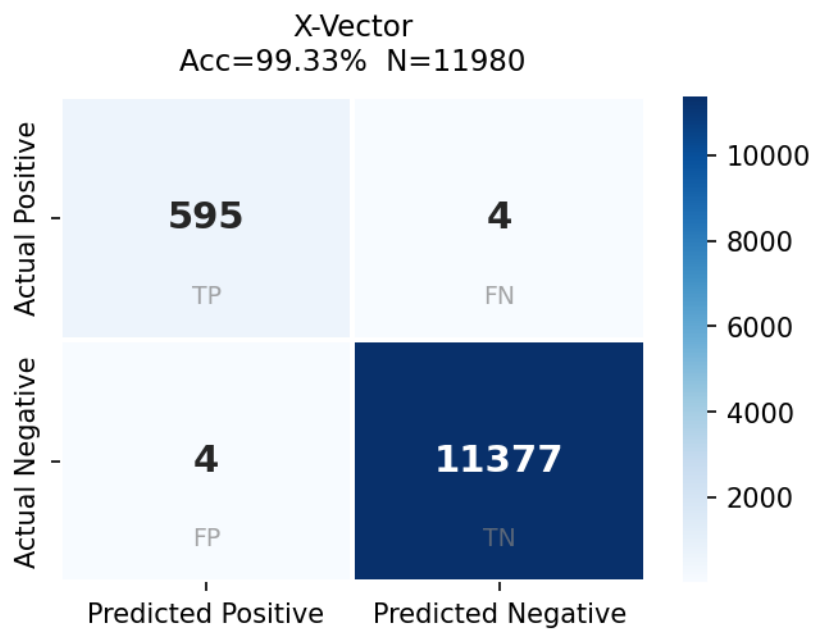


Figure 21: X-Vector Confusion Matrix

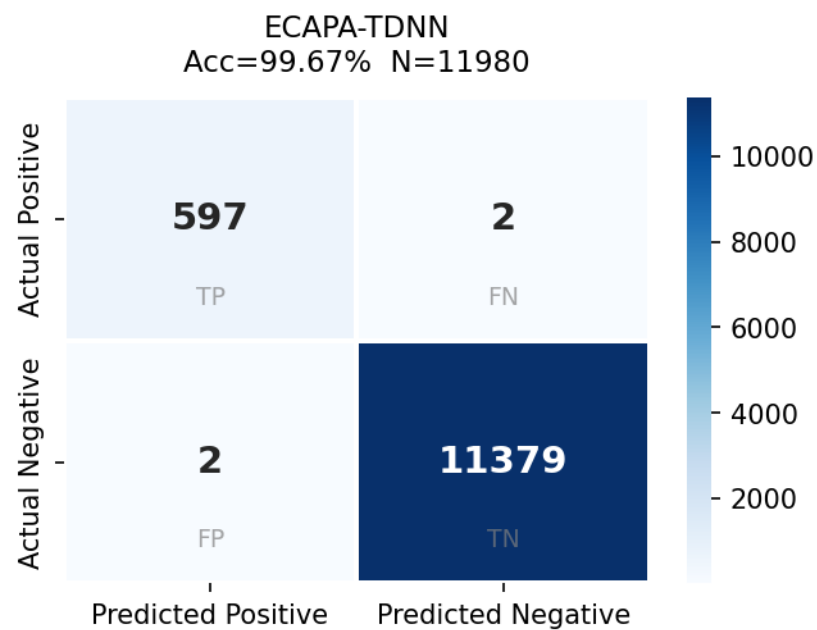


Figure 22: ECAPA-TDNN Confusion Matrix

Table 10: Speakers - Confusion Matrix Summary

Model	TP	TN	FP	FN
SVM (ML)	[593]	[11357]	[5]	[5]
Random Forest (ML)	[596]	[11360]	[2]	[2]
Gradient Boosting (ML)	[575]	[11339]	[23]	[23]
X-Vector (DL)	[595]	[11377]	[4]	[4]
ECAPA-TDNN (DL)	[597]	[11379]	[2]	[2]

7.3 Speaker-wise Confidence

Table 11: Per-Speaker Average Confidence Score by Model (%)

Speaker	SVM	Random Forest	Gradient Boosting	X-Vector	ECAPA-TDNN
Speaker_01	82.2%	53.9%	90.6%	99.8%	99.9%
Speaker_02	84.3%	60.5%	85.5%	99.9%	99.9%
Speaker_03	94.5%	81.8%	100.0%	97.6%	99.0%
Speaker_04	94.2%	71.0%	99.6%	100.0%	100.0%
Speaker_05	91.9%	60.2%	92.0%	100.0%	99.9%
Speaker_06	83.3%	67.7%	95.4%	99.9%	100.0%
Speaker_07	91.4%	87.8%	94.3%	97.5%	97.3%
Speaker_08	92.6%	68.6%	95.8%	98.9%	99.9%
Speaker_09	92.0%	63.9%	94.0%	100.0%	100.0%
Speaker_10	92.9%	82.3%	91.0%	100.0%	100.0%
Speaker_11	92.3%	66.9%	99.7%	99.9%	99.9%
Speaker_12	86.2%	61.4%	92.6%	99.8%	99.9%
Speaker_13	85.4%	55.8%	91.1%	100.0%	99.9%
Speaker_14	94.4%	83.5%	96.4%	99.9%	99.9%
Speaker_15	94.0%	66.5%	95.1%	99.8%	100.0%
Speaker_16	90.0%	81.9%	99.2%	99.9%	99.9%
Speaker_17	92.0%	61.3%	92.8%	97.2%	97.2%
Speaker_18	85.4%	54.0%	84.0%	96.0%	99.5%
Speaker_19	93.1%	66.2%	98.7%	100.0%	100.0%
Speaker_20	91.4%	73.8%	97.7%	99.8%	100.0%
AVERAGE	90.2%	68.5%	94.3%	99.3%	99.6%

7.4 Speaker-wise Precision / Recall / F1 Score Of Different Models

Table 12: Speaker Wise Summary of Precision of Different Models

Precision						
#	Speaker	SVM	Random Forest	Gradient Boosting	X-Vector	ECAPA-TDNN
1	Speaker_01	100.0%	100.0%	95.0%	100.0%	100.0%
2	Speaker_02	100.0%	100.0%	90.9%	95.7%	100.0%
3	Speaker_03	100.0%	100.0%	97.2%	100.0%	100.0%
4	Speaker_04	100.0%	100.0%	100.0%	100.0%	100.0%
5	Speaker_05	100.0%	100.0%	94.1%	100.0%	100.0%
6	Speaker_06	100.0%	100.0%	95.2%	95.5%	95.5%
7	Speaker_07	100.0%	100.0%	100.0%	100.0%	100.0%
8	Speaker_08	97.4%	97.4%	94.7%	97.4%	97.4%
9	Speaker_09	100.0%	100.0%	96.7%	100.0%	100.0%
10	Speaker_10	100.0%	100.0%	96.2%	100.0%	100.0%
11	Speaker_11	100.0%	100.0%	97.5%	100.0%	100.0%
12	Speaker_12	100.0%	100.0%	100.0%	100.0%	100.0%
13	Speaker_13	100.0%	100.0%	96.0%	100.0%	100.0%
14	Speaker_14	100.0%	100.0%	97.0%	100.0%	100.0%
15	Speaker_15	100.0%	100.0%	91.9%	100.0%	100.0%
16	Speaker_16	100.0%	100.0%	97.7%	100.0%	100.0%
17	Speaker_17	92.5%	97.4%	91.9%	97.3%	100.0%
18	Speaker_18	100.0%	100.0%	91.7%	100.0%	100.0%
19	Speaker_19	96.8%	100.0%	96.8%	100.0%	100.0%
20	Speaker_20	100.0%	100.0%	100.0%	100.0%	100.0%
	AVG	99.3%	99.7%	96.0%	99.3%	99.6%

Table 13: Speaker Wise Summary of Recall of Different Models

Recall						
#	Speaker	SVM	Random Forest	Gradient Boosting	X-Vector	ECAPA-TDNN
1	Speaker_01	100.0%	100.0%	100.0%	100.0%	100.0%
2	Speaker_02	100.0%	100.0%	90.9%	100.0%	100.0%

3	Speaker_03	100.0%	100.0%	100.0%	97.2%	100.0%
4	Speaker_04	100.0%	100.0%	100.0%	100.0%	100.0%
5	Speaker_05	100.0%	100.0%	94.1%	100.0%	100.0%
6	Speaker_06	90.5%	100.0%	95.2%	100.0%	100.0%
7	Speaker_07	97.1%	97.1%	94.3%	97.1%	97.1%
8	Speaker_08	100.0%	100.0%	97.3%	100.0%	100.0%
9	Speaker_09	100.0%	100.0%	96.7%	100.0%	100.0%
10	Speaker_10	100.0%	100.0%	92.6%	100.0%	100.0%
11	Speaker_11	100.0%	100.0%	100.0%	100.0%	100.0%
12	Speaker_12	100.0%	100.0%	93.3%	100.0%	100.0%
13	Speaker_13	96.2%	100.0%	92.3%	100.0%	100.0%
14	Speaker_14	100.0%	100.0%	97.0%	100.0%	100.0%
15	Speaker_15	100.0%	100.0%	94.4%	100.0%	100.0%
16	Speaker_16	100.0%	100.0%	100.0%	100.0%	100.0%
17	Speaker_17	100.0%	100.0%	91.9%	97.3%	97.3%
18	Speaker_18	96.0%	96.0%	88.0%	96.0%	100.0%
19	Speaker_19	100.0%	100.0%	100.0%	100.0%	100.0%
20	Speaker_20	100.0%	100.0%	100.0%	100.0%	100.0%
	AVG	99.0%	99.7%	95.9%	99.4%	99.7%

Table 14: Speaker Wise Summary of F1-Score of Different Models

F1-Score						
#	Speaker	SVM	Random Forest	Gradient Boosting	X-Vector	ECAPA-TDNN
1	Speaker_01	100.0%	100.0%	97.4%	100.0%	100.0%
2	Speaker_02	100.0%	100.0%	90.9%	97.8%	100.0%
3	Speaker_03	100.0%	100.0%	98.6%	98.6%	100.0%
4	Speaker_04	100.0%	100.0%	100.0%	100.0%	100.0%
5	Speaker_05	100.0%	100.0%	94.1%	100.0%	100.0%
6	Speaker_06	95.0%	100.0%	95.2%	97.7%	97.7%
7	Speaker_07	98.6%	98.6%	97.1%	98.6%	98.6%
8	Speaker_08	98.7%	98.7%	96.0%	98.7%	98.7%
9	Speaker_09	100.0%	100.0%	96.7%	100.0%	100.0%
10	Speaker_10	100.0%	100.0%	94.3%	100.0%	100.0%
11	Speaker_11	100.0%	100.0%	98.7%	100.0%	100.0%
12	Speaker_12	100.0%	100.0%	96.6%	100.0%	100.0%

13	Speaker_13	98.0%	100.0%	94.1%	100.0%	100.0%
14	Speaker_14	100.0%	100.0%	97.0%	100.0%	100.0%
15	Speaker_15	100.0%	100.0%	93.2%	100.0%	100.0%
16	Speaker_16	100.0%	100.0%	98.9%	100.0%	100.0%
17	Speaker_17	96.1%	98.7%	91.9%	97.3%	98.6%
18	Speaker_18	98.0%	98.0%	89.8%	98.0%	100.0%
19	Speaker_19	98.4%	100.0%	98.4%	100.0%	100.0%
20	Speaker_20	100.0%	100.0%	100.0%	100.0%	100.0%
	AVG	99.1%	99.7%	95.9%	99.3%	99.7%

CHAPTER 8:

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project successfully developed and implemented a speaker identification and verification system for the Assamese language by integrating both machine learning and deep learning techniques. The proposed framework followed a systematic workflow consisting of dataset collection, preprocessing, feature extraction, model implementation, performance evaluation, and real-time deployment through a Streamlit-based web application.

The preprocessing stage included speech annotation, silence removal, normalization, and resampling to ensure consistency across the recorded audio samples. For feature extraction, MFCC-based representations were utilized for machine learning models, while log-Mel spectrograms and speaker embeddings were employed for deep learning-based recognition.

The machine learning models, namely Support Vector Machine (SVM), Random Forest (RF), and Gradient Boosting (GB), demonstrated effective performance in speaker identification using MFCC features. In parallel, the deep learning approaches, X-Vector and ECAPA-TDNN, generated robust speaker embeddings that captured discriminative speaker characteristics and enabled reliable speaker verification through similarity-based comparison.

The system supports both closed-set speaker identification and speaker verification. Performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and confidence-based scoring mechanisms. The developed framework provides an efficient and practical solution for Assamese speaker recognition while remaining computationally feasible on standard hardware platforms.

Overall, this work contributes to the advancement of speech technology for Assamese, a low-resource language, and establishes a foundation for future research in speaker recognition and voice-based authentication systems.

8.2 Future Work

Although the proposed system achieved promising results, several enhancements can be explored in future research:

Open-Set Speaker Recognition: Extend the system to identify whether an input speaker is unknown and not present in the enrolled speaker database.

Unified Identification and Verification Framework: Enable both machine learning and deep learning models to perform speaker identification as well as speaker verification, allowing a comprehensive comparison of their capabilities.

Larger and More Diverse Dataset: Expand the Assamese speech corpus by including additional speakers, recording environments, age groups, and dialectal variations.

Real-Time Deployment: Optimize the system for deployment on mobile, embedded, and edge computing devices for practical real-world applications.

Multi-Language Extension: Extend the framework to support other low-resource languages of Northeast India, including Bodo, Mising, Karbi, and Dimasa.

References

- [1] D. A. Reynolds, T. F. Quatieri, and R. B. Dunn, "Speaker Verification Using Adapted Gaussian Mixture Models," *Digital Signal Processing*, vol. 10, no. 1–3, pp. 19–41, 2000.
- [2] T. Kinnunen and H. Li, "An Overview of Text-Independent Speaker Recognition: From Features to Supervectors," *Speech Communication*, vol. 52, no. 1, pp. 12–40, 2010.
- [3] R. Sriskandarajah and S. Sridharan, "Real-Time Speaker Recognition Using MFCC and Vector Quantisation on the TIDIGITS Corpus," in *Proc. International Conference on Spoken Language Processing (ICSLP)*, 1998.
- [4] H. S. Jayanna and S. R. Mahadeva Prasanna, "MFCC and Shifted MFCC with Vector Quantisation and Fuzzy Logic for Speaker Recognition," *International Journal of Computer Science and Network Security (IJCSNS)*, vol. 8, no. 3, pp. 213–220, 2008.
- [5] S. Kwon and S. Narayanan, "Emotion-Aware Speaker Recognition Using MFCC-SDC Features and a Multiclass Support Vector Machine," in *Proc. Interspeech*, pp. 1813–1816, 2007.
- [6] M. Ferras, C. Motlicek, H. Bourlard, et al., "Text-Dependent Speaker Recognition Using DWT and LPC on Algerian Berber Speech," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2010.
- [7] P. Boersma and D. Weenink, "Praat: Doing Phonetics by Computer," Version 6.x, Institute of Phonetic Sciences, University of Amsterdam, 2023.
- [8] M. Kachouri, R. Ejbali, and M. Zaied, "Speaker Identification Using Normalised Pitch Frequency and Mel-Frequency Cepstral Coefficients," *International Journal of Speech, Image and Video Processing (IJSPS)*, 2015.
- [9] D. Snyder, D. Garcia-Romero, G. Sell, D. Povey, and S. Khudanpur, "X-Vectors: Robust DNN Embeddings for Speaker Recognition," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5329–5333, 2018.
- [10] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification," in *Proc. Interspeech*, pp. 3830–3834, 2020.
- [11] F. Pedregosa, G. Varoquaux, A. Gramfort, et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

- [12] A. Ravanelli, T. Parcollet, P. Plantinga, et al., "SpeechBrain: A General-Purpose Speech Toolkit," arXiv preprint arXiv:2106.04624, 2021.
- [13] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [14] J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [15] C. Cortes and V. Vapnik, "Support-Vector Networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [16] A. Nagrani, J. S. Chung, and A. Zisserman, "VoxCeleb: A Large-Scale Speaker Identification Dataset," in *Proc. Interspeech*, pp. 2616–2620, 2017.
- [17] Z. Bai and X.-L. Zhang, "Speaker Recognition Based on Deep Learning: An Overview," *Neural Networks*, vol. 140, pp. 65–81, 2021.
- [18] S. Shyam and V. Srivastava, "Speaker Recognition for Low-Resource Indian Languages Using Transfer Learning," *International Journal of Speech Technology*, vol. 23, pp. 799–811, 2020.
- [19] Streamlit Inc., "Streamlit: An Open-Source Framework for Machine Learning and Data Science Applications," [Online]. Available: <https://streamlit.io>, 2023.
- [20] B. Goswami and P. K. Kalita, "Assamese Speaker Recognition Using an Artificial Neural Network Framework," in *Proc. National Conference on Signal Processing and Communication*, 2015.
- [21] D. Phukan and A. Sarma, "Speaker Identification Model for Assamese Language Using a Neural Framework," *International Journal of Computer Applications*, vol. 120, no. 8, 2015.
- [22] A. Paszke, S. Gross, F. Massa, et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [23] B. McFee, C. Raffel, D. Liang, et al., "librosa: Audio and Music Signal Analysis in Python," in *Proc. 14th Python in Science Conference (SciPy)*, pp. 18–25, 2015.
- [24] S. B. Davis and P. Mermelstein, "Comparison of Parametric Representations for Monosyllabic Word Recognition in Continuously Spoken Sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.

- [25] C. Li, X. Ma, B. Jiang, X. Li, X. Zhang, X. Liu, Y. Cao, A. Kannan, and Z. Zhu, "Deep Speaker: An End-to-End Neural Speaker Embedding System," arXiv preprint arXiv:1705.02304, 2017.
- [26] L. Wan, Q. Wang, A. Papir, and I. L. Moreno, "Generalized End-to-End Loss for Speaker Verification," in Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 4879–4883, 2018.
- [27] S. Hourri and J. Kharroubi, "A Novel Scoring Method Based on Distance Calculation for Similarity Measurement in Text-Independent Speaker Verification," *Procedia Computer Science*, vol. 148, pp. 256–265, 2019.
- [28] K. J. Devi and A. A. Devi, "Automatic Speaker Recognition Using MFCC and Artificial Neural Network," *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, vol. 9, no. 1S, pp. 39–42, 2019.

Turnitin Similarity Report



Page 2 of 66 - Integrity Overview

Submission ID trn:oid::3618:141236609

7% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- ▶ Bibliography
- ▶ Quoted Text
- ▶ Small Matches (less than 14 words)

Match Groups

-  **28 Not Cited or Quoted 6%**
Matches with neither in-text citation nor quotation marks
-  **3 Missing Quotations 1%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 5%  Internet sources
- 2%  Publications
- 4%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.